

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Summer Examination – 2025****Course:** Electrical Engineering**Branch:** Electrical Engineering /Electrical Engineering
(Electrical and Power)**Semester:** 6 (VI)**Subject Code & Name:** Industrial Automation and Control (BTEEE604A_Y19)**Max Marks:** 60**Date:** 08/07/25**Duration:****3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Automation Systems may include Control Systems but the reverse is not true. a. true b. False c. Not Applicable d. All	CO1	1
2	Automation also reduces _____ of production significantly by efficient usage of energy, manpower and material. a. Limit CO1 c. Oil d. Quality	CO1	1
3	The _____ elements translate the physical process signals such as temperature, pressure or displacement to convenient electrical or pneumatic forms of information. a. _____ CO1 c. sensing d. Generating Production	CO1	1
4	Sensors and Actuators Layer is _____ a. Level '0' CO1 c. Level '2' d. Level '3'	CO1	1
5	A modern controller device used extensively for sequence control today in transfer lines, robotics, process control, and many other automated systems is Called _____. a. Transfer CO1 c. Sensor d. Switching	CO1	1
6	The PLC is essentially a microprocessor-based real-time computing system. a. True CO1	CO1	1

7	The digital inputs modules convert the external binary signals from the process to the internal _____ signal level of programmable controllers a. Analogue CO1 c. Peak d. Low	CO1	1
8	Accuracy indicates the closeness of the measured value with the actual or true value. a. True CO1	CO1	1
9	Resistance thermometers employing metallic conductors for temperature measurement are called..... a. MTD CO1 c. Controllers d. Scale	CO1	1
10	One of below mentioned is not a type of Diaframgm. a. thin plate CO1 c. Corrugated d. thick Plate	CO1	1
11	Potentiometers are displacement sensors. They can be used for linear as well as displacement measurement. a. Triangular CO1 c. Temperature d. Angular	CO1	1
12	Which of the following is not an error in instrument readings a. Gross errors b. Systematic errors c. Temperature errors d. Random Error	CO1	1
Q. 2	Solve the following.		12
A)	Describe the hierarchical structure of Industrial Automation System.	CO2	6
B)	Draw the block diagrammatic structure of – i) An Industrial Sensor. ii) An Industrial Actuator and explain in very short.	CO2	6
Q.3	Solve the following.		12
A)	Explain the construction and principle operation of bourdon tube pressure gauge.	CO2	6
B)	Describe Various methods of Temperature measurements.	CO3	6
Q. 4	Solve Any Two of the following.		12
A)	Explain Components of Hydraulic Actuation Systems.	CO3	6
B)	Describe the types of automation systems.	CO3	6
C)	Explain RTD with help of neat diagram, the variation of resistance of metals with temperature with suitable expression and draw the graphs for commonly used materials.	CO3	6
Q.5	Solve Any Two of the following.		12

A)	Explain Industrial Actuator Systems with suitable block diagram	CO4	6
B)	Explain Industrial Sensors and Instrument Systems with Proper Block Diagram.	CO4	6
C)	Draw Architecture of PLCs and explain What is a communication Processor?	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	Explain with neat diagram electromagnetic type flow meter.	CO5	6
B)	Describe PID Controller in detail.	CO5	6
C)	Explain Components of Hydraulic Actuation Systems.	CO5	6
	*** End ***		